

ALPENA HARBOR, MI, USA

WMO: 997359

Lat: 45.0630N

Lon: 83.4286W

Elev: 587

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.8	(c) 2.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 25.2	(k) 28.3	(l) 22.0	(m) 29.1	(n) 6.1	(o) 320	(p) 0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 14.4	(c) 86.0	(d) N/A	(e) 82.2	(f) N/A	(g) 78.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.0	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.9	(b) 18.9	(c) 16.8	(e) -7.5	(f) 90.7	(g) 7.1	(h) 3.0	(i) -12.7	(j) 92.9	(k) -16.8	(l) 94.7	(m) -20.8	(n) 96.4	(o) -26.0	(p) 98.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	20.4	21.1	29.4	41.4	52.9	62.9	69.2	68.0	60.0	48.9	37.1	29.0
	DBStd	18.76	10.12	9.76	9.79	6.63	7.40	7.37	6.07	5.77	6.81	7.87	7.69	8.29
	HDD50	3847	917	810	639	271	50	1	0	0	4	116	389	650
	HDD65	7627	1382	1230	1103	709	383	124	27	31	181	506	836	1115
	CDD50	2078	0	0	1	12	139	388	594	558	303	80	3	0
	CDD65	385	0	0	0	0	8	61	156	124	30	6	0	0
	CDH74	2040	0	0	0	2	75	337	868	610	98	50	0	0
	CDH80	530	0	0	0	0	14	106	204	164	22	20	0	0
Wind	WSAvg	7.6	8.2	7.9	7.8	9.0	8.0	6.7	6.6	6.9	6.8	7.8	8.1	8.0
Precipitation	PrecAvg	28.5	1.7	1.3	2.0	2.2	2.9	2.8	2.9	3.2	3.0	2.2	2.2	1.8
	PrecMax	35.2	4.1	3.9	7.4	4.1	8.3	8.4	7.2	5.9	7.1	6.5	7.4	4.4
	PrecMin	20.2	0.2	0.1	0.4	1.2	1.0	0.2	0.2	0.9	0.3	0.6	0.6	0.4
	PrecStd	4.1	0.9	0.8	1.3	0.7	1.5	1.7	1.6	1.4	1.6	1.3	1.2	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	49.5	57.2	69.3	81.8	88.8	91.0	90.5	81.9	80.6	59.0	50.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	40.5	40.6	51.5	59.9	75.1	82.7	85.6	84.5	76.3	69.0	53.7	44.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	36.9	36.5	46.9	54.8	69.2	77.5	82.4	80.2	72.6	63.1	50.3	41.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	34.3	33.9	42.1	50.9	64.4	73.4	79.2	76.8	69.9	59.9	47.7	39.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	12.0	13.6	14.1	12.7	14.6	13.6	14.4	14.4	13.5	12.4	9.8	8.9
		MCDBR	12.0	14.3	18.5	19.5	22.7	18.3	18.1	18.5	16.8	17.4	13.7	10.5
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.285	0.298	0.308	0.352	0.373	0.405	0.403	0.392	0.370	0.341	0.310	0.282
	taud		2.438	2.371	2.449	2.373	2.389	2.323	2.348	2.376	2.430	2.489	2.522	2.493
	Ebn at Noon		265	282	295	288	284	274	273	272	269	263	255	256
	Edh at Noon		23	29	31	36	37	40	38	36	31	26	21	19
All-Sky Solar Radiation	RadAvg		397	687	1168	1474	1751	1914	1934	1641	1282	767	425	291
	RadStd		35	75	113	132	160	139	74	76	118	82	58	30

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page